The keys



White 1–16	the **16 pads** of bank **A** (or **B**): white 1 = pad 01, white 16 = pad 16
OCT+ **OCT–**	bank **B** / back to bank **A** (the LEDs A B show which)
black keys	**loops 1–11** (12–16: SEQ → LOOP): stopped at once; playing: from the end of the loop (its key blinks, the LCD says **LOOP 3 NEXT**)
ENV held + pads	**NOTE REPEAT**: the pads held hit on every step of AUTO CORRECT (OFF: 1/16), swung; recording, they go in. Stopped: at once, then in tempo. ENV tapped: tap tempo
ARP	PADS → **MULTI PITCH** (the last sound on all 27 keys, F4 as tuned) → **16 LEVELS** (white 1–16: the last sound, soft to loud) → PADS
MIDI in	notes 36–67 play A01–B16 (velocity in 16 steps); Program Change 0–15 picks loop 1–16

The factory pads

BANK A				BANK B			
13 CONGA	14 RIDE	15 SCRCH	16 COWBL	13 WOOD	14 CONGA +4	15 SCRCH −5	16 SNAP 33
09 TOM L	10 RIM	11 OHAT	12 SNAP	09 TOM L −5	10 RIM 33	11 SHAKR	12 TAMB
05 TOM H	06 CRASH	07 BASS	08 EP Dm9	05 CLAVE	06 CONGA −5	07 BASS −5	08 EP −7 (Gm9)
01 KICK	02 SNARE	03 HAT	04 CLAP	01 KICK 33	02 SNARE 33	03 HAT 33	04 CLAP 33

Pad 01 bottom left, as on the old sampling sequencers. **33**: TUNE −7 + SPEED 33, darker and grainier. Any pad, any of the 20 samples or your own: hit it, turn **PRESETS**. 12 bit at 40 kHz, pitched without interpolation.

The knobs

```
A01 KICK       090
CH1  CH2  CH3  CH4
100  100  100  100
```

KNOB 1–4	the **faders** of channels 1–4 (100 as set, 50 −12 dB); **SEL**: **5–8** (lit) / 1–4. The LCD's four columns always say what the knobs do
A page	EDIT, FX, SEQ, GLO open one (tap again: the next); untouched **12 s** (SETUP: BACK), or **HOME**: the pads and faders again
SELECT	tempo
ALGORITHM	the pad to edit (A01 … B16), without playing it
PRESETS	**the sample** of the selected pad: one sample on many pads, each with its own TUNE, DECAY …
MASTER	the volume

Pages · KNOB 1 · 2 · 3 · 4

EDIT WAVE	the pad's sample · COPY> pad · COPY (turn twice) · DELETE its own sample (turn twice)
SOUND	TUNE (semitones) · FINE · DECAY · LEVEL
TRUNC	START · END · DIR (fwd / rev) · SPEED (45 / 33)
OUT	CHOP group (--, 1–8: its pads cut each other) · PAN · CUT · RESO (only channels 1–2 filter)
SENDS	DRIVE · CHO · DLY · REV: **the effects of this pad**
FX FILTER	**DJ filter** on the mix: left low-pass, right high-pass, centre OFF · RESO
CHORUS	RATE · DEPTH · MIX
DELAY	TIME (1/4 … 16T, in tempo) · FDBK · COLOR · MIX
REVERB	SIZE · DAMP · PRE (ms)
SEQ LOOP	LOOP · BARS (AUTO, 1–32) · QUANT (AUTO CORRECT, also NOTE REPEAT's grid) · SWING 50–71 %
LOOP TOOLS	CLEAR (turn twice) · COPY> loop · COPY (turn twice)
SONG	STEP · SEG (its loop) · REPEAT (0: the song ends here) · SONG OFF / 1–4
GLO SETUP	TEMPO · BACK · – · RESET (**all loops gone**, turn twice); tap: CLICK: CLICK · COUNT · DUB; tap: OUTPUT: SMOOTH; tap: **SAMPLE**

Make a beat

1. **REC** (**ARMED**), **PLAY**: **COUNT 4 3 2 1**, then it records.
2. Kick (white 1) and snare (white 2). **AUTO CORRECT** puts them on the grid.
3. **REC** again at the end: an empty loop (BARS **AUTO**) is now as long as your take.
4. Hats with **NOTE REPEAT**: **REC**, hold **ENV** and white 3: 1/16 hats while you hold. QUANT 1/8 or 1/32 for other rolls, SWING for the feel.
5. Or step by step: hit the hat, **hold** **SEQ**: the white keys are its 16 steps; **OCT+** / **OCT–** the next / last bar.
6. Ghost notes: **ARP** twice (**16 LEVELS**), white 1–16 play the snare soft to loud.
7. Another loop: a black key, record into it. Songs: SEQ → SONG.

Buttons

PLAY	run / stop
REC	stopped: ARMED; playing: overdub on / off; **hold 2 s**: clear the loop
ENV	tapped: tap tempo · held + pads: **NOTE REPEAT**
HOME	from a page: the pads · on the pads: **TRACK MUTE** on / off (HOME blinks): white keys mute / unmute their pad, **ENV** held + pad: **solo**; muted pads stay muted when you leave
GLO held + white	1–8 **mute**, 9–16 **solo** ch 1–8
EDIT held + OCT–	**undo** the last clear / erase / copy (again: redo)
FX held + white 1–4	**punch-in**: roll 1/8 · roll 1/16 · reverse · tape stop; **OCT–** / **OCT+**: reverb / echo throw
LFO + pad	**erase** its hits as the playhead passes (stopped: all); **LFO** tapped: **SAMPLE**
SAVE	save now; **held + a black key**: the loop into that loop

Channels

8 channels, one sound each: a pad plays on **its place in the row pair**: pads 01 and 09 on channel 1, 02 and 10 on 2 … 08 and 16 on 8 (in both banks). **A new hit on a channel cuts the last**: HAT (03) and OHAT (11) choke. Channels 1–2 (pads 01 02 09 10) have the resonant filter, 3–6 a fixed low-pass, 7–8 none.

Based on zp12, SLOOP (isod89) and Felucca (Leo Kuroshita, Hügelton Instruments), GPL-3.0 · sounds CC0 (VSCO-2 CE, Sonic Pi), E-piano Greg Sullivan (CC BY 3.0)

Inspired by the sampling sequencers of the late 80s, names their owners' · not affiliated with M-VAVE or Akai

What the lights say

PLAY	lit while it runs
REC	lit: recording · blinking: ARMED, or an overdub waiting for the next 1
ENV	blinks on the beat (tap along)
ARP · SEL	MULTI PITCH or 16 LEVELS on · the knobs are faders 5–8
LFO	erasing (held + a pad)
SAVE	lit in song mode
OCT− OCT+	bank A / B on the keys
HOME	blinking: TRACK MUTE; the white keys lit while their pad is heard, glowing when muted
White keys	flash when their pad sounds; **SEQ held**: the steps with a hit; **GLO held**: 1–8 the channels you hear, 9–16 the soloed
Black keys	the loop playing lit, the next one blinking

The screen

Big line	the selected pad and its sound (A01 KICK), or what just changed
Top right	the tempo, with **ARMED**, **COUNT** or **REC** before it; below it the page
4 × 4 pads	the bank on the keys, pad 01 bottom left; red while it sounds, the selected one outlined, **greyed out with a red bar: muted**, yellow ring: soloed
Faders	beside the pads: channels 1–8, the four on the knobs white; M muted, S soloed
A B	the bank on the keys
M · L	MULTI PITCH · 16 LEVELS
S · NR	song mode · NOTE REPEAT running
RUN · REC	running (blinks on the beat) · recording (blinks: ARMED)
Red dot	in the step grid: a hit with its own TUNE / FINE / DECAY / CUT

Recording

Count-in	GLO → CLICK → COUNT: OFF, 1 bar (4 3 2 1), 2 bars
AUTO	a loop with BARS AUTO takes the length of its first take, in whole bars
QUANT	AUTO CORRECT: OFF, 1/8, 1/8T, **1/16**, 1/16T, 1/32, 1/32T
SWING	50, 54, 58, 63, 67, 71 %: the off 1/16s (QUANT 1/8: the off 1/8s) come later; NOTE REPEAT swings along
NOTE REPEAT	its grid is QUANT (OFF: 1/16); playing, the first hit waits for the grid; every repeat is recorded like a hit
Velocity	16 steps. The keys play at one level (14 of 16), **16 LEVELS** gives all 16, MIDI in its own
Overdub	a hit where one is: the louder stays. Up to 512 hits a loop
Per hit	TUNE, FINE, DECAY, CUT turned while recording stay with the pad's hits
Step grid	16 steps of 1/16 a bar; a step set by hand plays at level 14

Loops and songs

16 loops	1–11 on the black keys, 12–16 with SEQ → LOOP; each 1–32 bars or AUTO
Copy	SAVE held + a black key, or SEQ → TOOLS
Songs	4 songs of up to 32 steps, each a loop (SEG) played REPEAT times; PLAY plays it from step 1

Sampling over USB

1. Computer or phone: sound output **zpc60 FM-1** (Linux: "pid.codes Test PID"), play.
2. Hit the pad, tap LFO: the meter moves. RATE 40/26/20K · TIME · THRESH · MON.
3. REC: ERASE %, then REC; REC again or TIME ends it. On the pad as SMP01–24; again on that pad: replaced.

Ranges

TEMPO	40–240 BPM; SELECT 0.5 a click, ENV tapped
Faders	0–127: 100 as set, 50 −12 dB
TUNE · FINE	−24 … +12 semitones · ±50 cents
Samples	the kit 12 bit at 40 kHz; your own at 40, 26 or 20 kHz (less rate, more time)
Filters	channels 1–2: CUT, RESO (4-pole low-pass) · 3–6: fixed low-pass · 7–8: none

Saving and safety

Saved	by itself when stopped and 3 s after a change; SAVE now
Its own	zpc60 keeps its own save; it never reads zp12's (they share the flash: switching back and forth, the last one saving wins)
Not saved	TRACK MUTE, mute / solo and the DJ filter
Undo	EDIT held + OCT−: the last clear, copy or stopped erase
RESET	GLO → SETUP: everything back to the factory, **all loops gone**
Rescue	no answer? Hold OCT− while switching on: the USB rescue, then install again

GPL-3.0 · github.com/zvenson/zpc60